SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
 <input type="text" name="name">
 <input type="email" name="email">
 <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate HTML5 semantic elements missing in the structure.

Missing: <!DOCTYPE html>, <html lang>, <head> (with <meta charset> and <title>), <body>, <header>, <main>, <section>, <fieldset>/<legend>, <label> for each input, and <footer>. The button also has no type attribute.

2. Modify the structure to make it standards-compliant.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Symposium Registration</title>
</head>
<body>
  <header><h1>Online Symposium Registration</h1></header>
  <main>
    <section>
      <h2>Register Now</h2>
      <form action="/register" method="POST">
        <fieldset>
          <legend>Student Details</legend>
          <label for="name">Name:</label>
          <input type="text" id="name" name="name" required>
          <label for="email">Email:</label>
          <input type="email" id="email" name="email" required>
          <button type="submit">Submit</button>
        </fieldset>
      </form>
    </section>
  </main>
```

```
  <footer><p>&copy; 2026 College Symposium</p></footer>
</body>
</html>
```

3. Interpret the HTTP request generated during form submission.

The browser builds a POST request with the data in the body, not the URL:

```
POST /register HTTP/1.1
Host: college.edu
Content-Type: application/x-www-form-urlencoded
Content-Length: 38

name=Arun+Kumar&email=arun%40mail.com
```

The request line holds the method, path and version. The headers describe the host, content type and length. The body carries the URL-encoded name=value pairs.

4. Analyze the HTTP response cycle from server to browser.

(a) The browser resolves the domain through DNS and opens a TCP connection (plus a TLS handshake for HTTPS). (b) It sends the POST request. (c) The server processes it: validates the data, stores it in the database and generates a result. (d) The server returns a status line and headers (e.g. HTTP/1.1 200 OK, or 302 Found with Location: /success), followed by the HTML body. (e) The browser parses the HTML, requests any CSS, JS or images, and renders the page. For a 302, the browser makes a new GET to the redirect URL (the Post/Redirect/Get pattern, which prevents duplicate submissions on refresh).

5. Suggest improvements for usability and accessibility.

Add <label for> on every field, placeholder and required attributes, autocomplete hints, and a clear success or error message. Use sufficient color contrast, keyboard-friendly tab order, aria-describedby for error text, and a responsive viewport meta tag. Add client-side validation (pattern, minlength) as well as server-side validation, and use HTTPS.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
 <head>
 <title>Company</title>
 </head>
 <body>
 <div>
 <h1>Welcome
 <p>About us
 </div>
 </body>
</html>
```

Questions:

1. Identify improper nesting and missing closing tags.

<h1> and <p> are never closed, and the <p> ends up inside the unclosed <h1>, which is invalid because headings cannot contain block-level paragraphs. There is also no <!DOCTYPE>, no lang attribute and no <meta charset>.

2. Analyze how different browsers may interpret this structure.

Without a DOCTYPE, browsers switch to quirks mode, which changes box-model and spacing behavior. HTML parsers then error-correct differently: some auto-close <h1> when they meet <p>, while others keep <p> inside the heading, so the DOM tree differs, and so do default margins and CSS inheritance. That produces the inconsistent layout.

3. Identify missing semantic elements (e.g., header, section).

<header>, <main>, <section> (for About Us), <footer> and optionally <nav>. The generic should be replaced by these.

4. Provide a corrected W3C-compliant HTML structure.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Company</title>
</head>
<body>
  <header>
    <h1>Welcome</h1>
  </header>
  <main>
    <section>
      <h2>About Us</h2>
      <p>About us</p>
    </section>
  </main>
  <footer><p>&copy; 2026 Company</p></footer>
</body>
</html>
```

5. Suggest techniques to ensure cross-browser compatibility.

Always declare the HTML5 DOCTYPE. Validate with the W3C Validator. Use a CSS reset or normalize.css. Use feature detection (@supports, Modernizr) rather than browser sniffing. Add vendor prefixes (autoprefixer) and polyfills for old browsers. Use responsive units, flexbox and grid with fallbacks, and test on Chrome, Firefox, Safari and Edge (BrowserStack or DevTools device mode). Apply progressive enhancement and graceful degradation.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
 <input type="text" name="username">
 <input type="password" name="password">
 <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the form configuration.

(a) <button type="button"> is a plain button that does nothing, so it never submits the form. (b) method="GET" sends the username and password in the URL, which is insecure. (c) There are no <label> elements, no ids, no required and no autocomplete. (d) The password is not protected by HTTPS (the action is a relative path, so it depends on the page's protocol).

2. Analyze why the HTTP request is not triggered.

A submit event fires only when a type="submit" button (or an <input type="submit">) is clicked, or Enter is pressed in a field. type="button" has no default action, and no JavaScript handler such as form.submit() is attached, so the browser never builds or sends a request.

3. Interpret the role of GET method in this scenario.

Even once fixed, GET appends the data to the URL (/submit?username=abc&password=1234). Credentials then appear in the address bar, browser history, server logs, proxy logs and Referer headers. GET is meant for safe, idempotent data retrieval (searches, filters), not for sending credentials or changing state.

4. Propose a corrected implementation.

```
<form action="/submit" method="POST">
  <label for="username">Username:</label>
  <input type="text" id="username" name="username" required autocomplete="username">
  <label for="password">Password:</label>
  <input type="password" id="password" name="password" required minlength="8"
autocomplete="current-password">
  <button type="submit">Login</button>
</form>
```

5. Suggest improvements for secure data transmission.

Serve the site over HTTPS/TLS and use HSTS. Use POST for credentials and never put them in the URL. Hash passwords on the server (bcrypt or Argon2) and never store plain text. Add CSRF tokens, server-side validation and sanitization, rate limiting and account lockout, and secure HttpOnly, Secure, SameSite cookies. Optionally add multi-factor authentication and a CAPTCHA.

Code of Conduct:

I P. Lohith Krishna (192325122) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Lohith Krishna

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand